Supplemental Material for
"Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$"

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Contents

S1 The SU(3) qutrit and why its classical dynamics is quantum-exact

Tm^{3+} is $4f^{12}$, ${}^3H_6$ ($J = 6$). In the $C_s(m)$ crystal field of the $Pbnm$ $4c$ site the 13-fold multiplet splits into 13 singlets; the model keeps the three lowest, $|1\rangle, |2\rangle, |3\rangle$, with splittings $E_{12} = 2.068\,\mathrm{meV}$ ($0.50\,\mathrm{THz}$) and $E_{13} = 4.963\,\mathrm{meV}$ ($1.20\,\mathrm{THz}$), consistent with the measured lines $E_{12} + E_{23} \approx E_{13}$. The site state is the Gell-Mann vector $\lambda^a = \mathrm{Tr}(\rho\Lambda^a)$, $a = 1\ldots8$, and every single-ion term (CEF energies, external drives, exchange mean fields) is *linear* in the Λ^a. For a Hamiltonian linear in the generators the von Neumann equation $\dot\rho = -i[H,\rho]$ closes on $\vec\lambda$:

$$\dot\lambda^a \;=\; \Omega_{ab}(t)\,\lambda^b - \gamma_a\big(\lambda^a - \lambda^a_{\rm eq}\big), \qquad \Omega_{ab} = \tfrac{1}{2}\,\mathrm{Tr}\Big(\Lambda^a\,(-i)[H(t),\Lambda^b]\Big)\,. \tag{1}$$

The classical SU(3) propagation is therefore the *exact* quantum dynamics of the driven, damped three-level ion, and a Boltzmann-weighted ensemble of trajectories launched from the level seeds $|1\rangle, |2\rangle, |3\rangle$ is the exact thermal density matrix for single-ion drives. (This was verified explicitly against an independent density-matrix integrator; the two censuses agree to all digits quoted.) Damping is Bloch-type on the Gell-Mann pairs of each transition, $\gamma(\lambda^{1,2}), \gamma(\lambda^{4,5}), \gamma(\lambda^{6,7})$, with population relaxation $\gamma(\lambda^{3,8})$ toward the equilibrium populations captured at initialization.

The magnetically bright coordinate of the E_{12} block follows from the projected moment $PJ_zP = 5.264\,\lambda^2$: only λ^2 radiates magnetically, and the onsite splitting drives the planar precession $\dot\lambda^1 = -\Delta_{12}\lambda^2$, $\dot\lambda^2 = +\Delta_{12}\lambda^1$, so the E_{12} sector is a single driven oscillator

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t-t')]\,h^1(t')\,dt', \tag{2}$$

where h^1 is the effective force a conversion vertex exerts on λ^1. All delay-axis structure of a cross peak lives in the τ-dependence of h^1.

S2 Crystallography and the projected Fe–Tm coupling

$TmFeO_3$ is *Pbnm* ($Z = 4$): Fe^{3+} at 4*b* (site symmetry $\bar 1$; every Fe sits on an inversion center), Tm^{3+} at 4*c* (site symmetry the single mirror m_z). Inversion maps each Fe sublattice to itself and exchanges Tm sublattices in pairs; the local mirror is the physical 4*c* site mirror and generates the relative λ^5, λ^7 signs between Tm sublattices. The Fe order is G-type ($T_N \approx 632\,$K) with DM canting; below the spin reorientation ($T_2 \approx 85\,$K) the ground state is $\Gamma_2 = (F_x, C_y, G_z)$.

The Fe–Tm sector is built from the physical operator route

$$\boldsymbol S,\ \boldsymbol J \;\longrightarrow\; H_{\rm Fe-Tm} = \boldsymbol S^T A\,\boldsymbol J \;\longrightarrow\; P\boldsymbol J P \;\longrightarrow\; \text{couplings to } \lambda^a, \tag{3}$$

never by assuming that a space-group rotation acts as a closed onsite SU(3) rotation in one fixed qutrit basis. Working in a *transported gauge* (qutrit bases on the four Tm sublattices related by the group operations themselves) makes every bond formula explicit and turns the irrep (Γ_1) projection into simple four-site sign sums. Time-reversal splits the Gell-Mann set into $\lambda^- = \{\lambda^2, \lambda^5, \lambda^7\}$ (odd) and $\lambda^+ = \{\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8\}$ (even); every coupling term must be $\mathcal{T}$-even overall, which fixes how many Fe-spin or B-field legs may attach to each sector. The candidate couplings through cubic order in the fields are

family	operator form	sector	legs
K^-	$S_\alpha\lambda^a$	λ^-	1 Fe
ESJ	$E_\eta S_\alpha\lambda^a$	λ^-	1 E-photon + 1 Fe
κ^B	$B_\eta S_\alpha\lambda^a$	λ^+	1 B-photon + 1 Fe
W	$S_\alpha S_\beta\lambda^a$	λ^+	2 Fe

The full orbit formulas (all eight coefficients per bond, all four Tm sublattices, both time-reversal sectors) and the implementation prescriptions for global-axis versus local-frame storage are given in the long-form working note that accompanies the data (`tmfeo3_foundation.tex`, Secs. 3–14).

S3 The sharp requirement and the vertex scorecard

Two collinear pulses give $M_{\rm NL} = M_{AB} - M_A - M_B$; order by order only the mixed terms survive, so $M_{\rm NL}$ starts at second order and the leading robust magnetic-dipole contribution is third order. Combining Eq. (2) with the conservation rules at each vertex (energy; $\mathcal{T}$-parity; Γ_1) yields six requirements for a coupling to produce the geometry-A cross peak at $(q_{\rm AFM}, E_{12})$:

R1 (detection) reaches λ^1/λ^2 (the E_{12} block);

R2 (drive) is sourced by the pumped δS_y magnon;

R3 (parity) is $\mathcal{T}$-even (fixes the $\lambda^{\pm}$ sector);

R4 (irrep) is Γ_1-allowed in the Γ_2 state;

R5 (energy) carries the $0.90 \to 0.50\,$THz mismatch (needs ≥ 2 non-λ legs);

R6 ($M_{\rm NL}$) involves both pulses.

Table S1: Vertex scorecard. Only W and κ^B pass all six requirements.

vertex	legs	sector	R1	R2–R6	verdict
K^-	$1\text{Fe} + 1\text{Tm}^-$	λ^-	$\times$	$\times$	hybridizes only; Γ_1-blocked; reorients Γ_2
ESJ	$E + \text{Fe} + \text{Tm}^-$	λ^-	$\times$	✓	real, but feeds E_{13}/E_{23}, not E_{12}
W	$2\text{Fe} + \text{Tm}^+$	λ^+	✓	✓	**winner**: rank-1, factorized peak
κ^B	$B + \text{Fe} + \text{Tm}^+$	λ^+	✓	✓	works; B_z-gated $\Rightarrow$ geometry-B vertex

K^- fails threefold. (i) Linear in S, it only rotates the harmonic normal-mode basis: coupled harmonic oscillators give diagonal peaks only; a static bilinear conserves energy and cannot convert 0.90 into 0.50 THz. (ii) The Γ_1 orbit sums vanish for every channel except K^-_{2y}: a static one-at-a-time scan at 0.2 meV leaves the energy, the Fe order, and the uniform $(\lambda^2, \lambda^5, \lambda^7)$ *bit-identical* to baseline for all components except $2y$. (iii) The one allowed channel K^-_{2y} is a transverse molecular field on the order parameter and reorients $\Gamma_2 \to G_y$ between 0.085 and 0.090 meV—it is not a perturbative vertex. Full-dynamics 2DCS on the inert channels (K^-_{2x} up to 1.5 meV, K^-_{2z} up to 2.0 meV) leaves the induced λ^2 at machine precision ($\leq 7 \times 10^{-15}$).

The two winners. The two-magnon vertex rectifies the pulse-A/pulse-B cross term,

$$h^1_{\rm NL}(t,\tau) = W_1^{yy} A^2 \left[\cos(q_{\rm AFM}\tau) - \cos(q_{\rm AFM}(2t+\tau))\right]\theta(t) \;\Rightarrow\; M^W_{\rm NL} \propto \cos(q_{\rm AFM}\tau)\left[1 - \cos\omega_{12} t\right], \quad (4)$$

a clean factorized peak with no diagonal partner. The field-gated vertex kicks at the probe, $h^1 \propto B_y(t) S_y(t)$ with $S_y(0) = A \sin q_{\rm AFM}\tau$, giving $M^\kappa_{\rm NL} \propto \sin(q_{\rm AFM}\tau)[1 - \cos\omega_{12} t]$: also on target, but the same gate dumps each pulse's local response into Tm at both pulse times, generating an (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background. Its dominant symmetry slice is z-gated ($B_z(t)$), so in geometry A ($B_z = 0$) it is *identically silent*: the gate is simultaneously the reason it is harmless in geometry A and active in geometry B.

Exhaustion of the quadratic families. All remaining symmetry-allowed quadratic couplings were switched on one at a time in the full nonlinear dynamics and are exactly inert (differences at the 10^{-12} level or below): $W_4^{yy,xy,yz,xx,zz,xz}$, W_1^{xy}, $W_6^{yy,xz}$, κ^E_{2y}, κ^E_{5x}. Among the W_1 components: W_1^{yz} inert, W_1^{zz} negligible, W_1^{xx} alive but floods the spectrum with an $\omega_t = 0.16\,\mathrm{THz}$ comb, W_1^{xz} alive and on-target for hybridization (used below), and no static component can replace the gated κ^B for the geometry-B transfer peak: the entire W_1+W_4 family fails to produce $(E_{12}, q_{\mathrm{AFM}})$ in geometry B without simultaneously contaminating geometry A. The B_z-gate is therefore *proven necessary by exhaustion.*

S4 Numerical protocol

Dynamics. Classical LLG for Fe (unit spins, $S = 5/2$ scale absorbed), SU(3) Bloch dynamics for the four Tm sublattices; RK4 with $\Delta t = 0.02$ code units; $1 \times 1 \times 1$ cell of the $Pbnm$ magnetic unit (8 Fe + 4 Tm). Time is measured in $\hbar$/meV; a frequency f in THz corresponds to $E = f \times 4.136\,\mathrm{meV}$.

2DCS protocol. Delay scan $\tau \in [-120, 0]$ code units, $\Delta\tau = 0.1$ (0.2 for scans); detection window to $t_{\max} = 160$ for 0.01 THz resolution on ω_t. The nonlinear signal is the honest three-run subtraction $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$ per delay; no pulse-window chunking; no reuse of the reference run across delays (both shortcuts were found to contaminate spectra and are documented pitfalls). The pump enters as a tabulated waveform (the measured field, times converted $t_{\mathrm{code}} = t_{\mathrm{ps}}/0.6582$, auto-centered and normalized).

Spectra and blind peak census. 2D spectra are $|\mathrm{FFT}_{\tau,t}|$ of M_{NL} with: detection window $t \geq 3$ (excludes pulse overlap), Gaussian t-apodization to 0.03, cosine tapers on both τ edges, global mean subtraction. The delay axis is plotted physically ($\omega_T = -\omega_\tau^{\mathrm{FFT}}$; $+ =$ non-rephasing). Peaks are accepted only by a blind census: 2D local maxima (maximum filter, 0.10 THz neighborhood) with prominence ≥ 1.5 over the local median background (0.34 THz annulus) and amplitude $\geq$ 4–5% of the map maximum, merged within 0.09 THz. Rectified ($\omega_t \approx 0$) content is analyzed separately via $S_{\mathrm{DC}}(\tau) = \langle M_{\mathrm{NL}} \rangle_t$ with cubic drift removal.

Thermal ensembles. Boltzmann mixtures of trajectories from the level seeds $|1\rangle, |2\rangle, |3\rangle$ (Sec. S1). Two documented pitfalls: (i) any pre-dynamics energy minimization collapses excited-level seeds and must be disabled; (ii) population damping $\gamma(\lambda^{3,8})$ relaxes toward the equilibrium captured at initialization, which must therefore be captured per-seed.

S5 Experimental inputs

Two consequences organize the readout physics. First, excitation scales as E^2 and detection as E^1 (third order). Second, the CEF lines are E1-bright while q_{AFM} is E1-dark but M1-bright: a magnon is therefore read out either by borrowing a CEF electric dipole through an Fe–Tm vertex (geometry A) or through its own magnetic dipole at the magnon frequency (geometry B). Using the magnetic g-factor instead of $\sqrt{f}$ for the magnon electric emission over-weights it by $\sim 100\times$ and buries the CEF peaks under the magnon diagonal; the oscillator-strength weighting resolves this without any tuning.

Table S2: Measured mode parameters and the derived simulation inputs. CEF damping is Bloch on the Gell-Mann pair of each transition, $\gamma_a[\mathrm{meV}] = \gamma[\mathrm{THz}] \times h$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$ fixed by the q_{AFM} line. Emission weight is $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons. The last column is the measured pulse spectral amplitude at each mode.

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
q_{FM}	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
q_{AFM}	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157\,\mathrm{meV}$	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414\,\mathrm{meV}$	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207\,\mathrm{meV}$	0.36

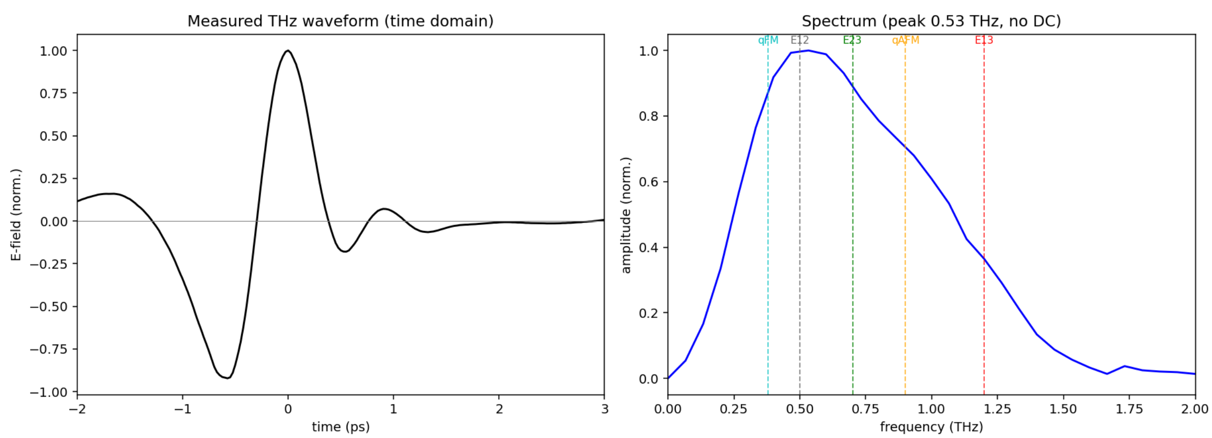

Figure S1: The measured THz pump. Left: single bipolar cycle ($\sim 1.2\,$ps, no DC). Right: amplitude spectrum peaked at $0.53\,\mathrm{THz} \approx E_{12}$, covering all five modes. The absence of DC content removes the quasistatic-tilt legs that a Gaussian stand-in pulse would add; the measured waveform is the cleanest configuration.

S6 Displacive verification of the W mechanism

The rectified two-magnon force is a suddenly switched step: its edge carries spectral weight $\propto e^{-\omega_{12}^2\sigma^2/2}$ at the CEF frequency, i.e. the E_{12} quantum is drawn from the pulse bandwidth (intra-pulse difference-frequency/impulsive-Raman), while the magnon pair carries only the phase memory $\cos q_{\mathrm{AFM}}\tau$. Verified directly: stretching the pulse $\sigma = 0.25 \to 1.0$ at fixed amplitude collapses the peak by 4.5 decades ($51.7 \to 1.7 \times 10^{-3}$), while switching the direct Tm drive on/off changes it by $< 10^{-4}$ (pure-W origin). The same sweep resolves the second Fe leg: impulsive pulses use the resonant probe magnon (peak $\propto A^2$); longer pulses cross over to the field-following quasistatic tilt (peak $\propto A^1$). The measured no-DC waveform suppresses the quasistatic legs, which is why the experimental pulse yields the cleanest spectra.

S7 Geometry B: gate versus buildup

Isolated-vertex runs: κ_{1y}^B alone $\Rightarrow (E_{12}, q_{\mathrm{AFM}})$ present; W_1^{xz} alone $\Rightarrow$ peak absent. In the consolidated Hamiltonian $W_1^{yy} = W_1^{xz} = 0.01$ and $\kappa_{1y}^B = 0.0035$: the geometry-B peak ratio is the observable that calibrates κ^B, and because κ^B is B_z-gated the recalibration provably cannot affect either $H \parallel a$ channel (verified bit-identical).

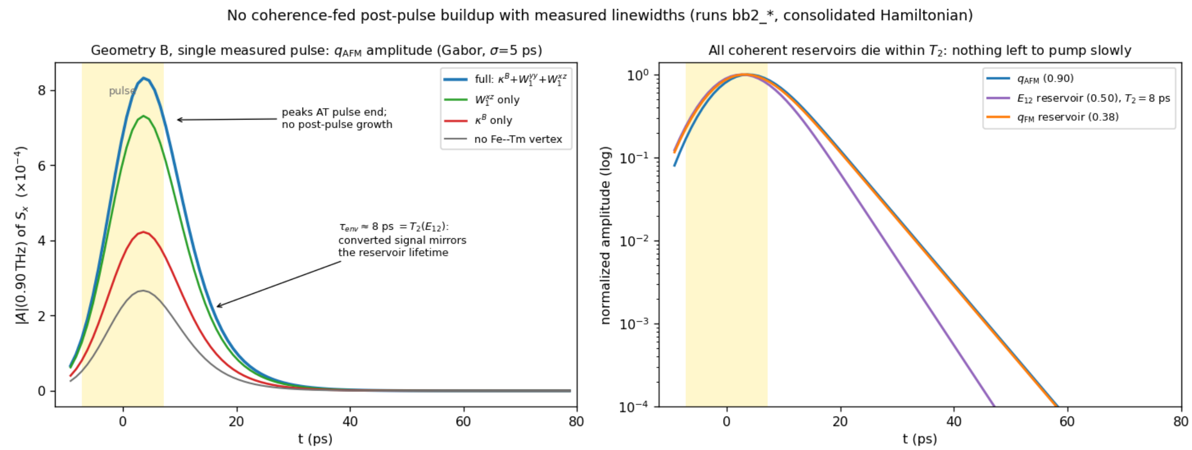

Figure S2: No coherence-fed post-pulse buildup once the measured linewidths are enforced (single measured pulse, geometry B, runs `bb2_*`; Gabor amplitude, $\sigma = 5\,\mathrm{ps}$, artifact-free local analysis). Left: the q_{AFM} amplitude for the four vertex configurations peaks *at* pulse end and decays with $\tau_{\mathrm{env}} \approx 8\,\mathrm{ps} = T_2(E_{12})$; the vertices set the absolute scale but none produces post-pulse growth. Right: all coherent reservoirs die within their measured T_2 (log scale)—there is nothing left to pump the magnon slowly.

The post-drive buildup: a corrected analysis. An earlier stage of this project attributed the experimentally reported post-pulse growth of the geometry-B signal to slow W_1^{xz}-mediated exchange through the nearly closed triad $q_{\mathrm{FM}} + E_{12} = 0.88 \approx q_{\mathrm{AFM}}$. That attribution does not survive the enforcement of the measured linewidths. Because $B \parallel c$ never drives q_{AFM} directly, *all* geometry-B magnon amplitude is converted, and a converted signal inherits its source's lifetime: with $T_2(E_{12}) = 8\,\mathrm{ps}$ the E_{12}-fed component (and with it the triad) is dead within $\sim$10 ps. The artifact-free Gabor analysis of Fig. S2 shows the q_{AFM} amplitude peaking at pulse end and decaying monotonically in every vertex configuration; the earlier "buildup" was visible only while $\lambda^{1,2}$ were left undamped (an infinitely lived E_{12} reservoir, inconsistent with the measured 0.038 THz linewidth). W_1^{xz} therefore loses its buildup role—but it acquires a direct observable of its own instead (next paragraph); it remains bounded above ($\lesssim 0.02$) by the mixer sweep of the preceding paragraph.

The reverse transfer peak: W_1^{xz}'s own observable. The experiment reports an additional geometry-A *cross*-channel peak near (0.5, 1.0). The model contains a genuine local maximum at exactly $(+0.49, 0.90) = (E_{12}, q_{\mathrm{AFM}})$ in the λ^2 channel (0.063 of the main peak; run `vA`), with a weaker $2E_{12} = 1.00\,\mathrm{THz}$ harmonic companion at 0.021. Isolated-vertex tests identify the mechanism: with $W_1^{xz} \to 0$ the peak collapses sevenfold ($0.063 \to 0.009$; run `vA_noWxz`) while the main $(q_{\mathrm{AFM}}, E_{12})$ peak is untouched, and doubling the Tm drive doubles it ($0.063 \to 0.123$; run `vA_su3x2`), i.e. it is linear in the stored E_{12} amplitude. This is the *reverse* of the main transfer: the pump-stored E_{12} coherence is converted *into* the q_{AFM} magnon by the static hybridization—the static W family read in both directions, magnon→CEF through W_1^{yy} and CEF→magnon through W_1^{xz}. The measured amplitude ratio of the reverse to the forward peak calibrates W_1^{xz} directly. Assignment discriminators for the experimental peak: an $(E_{12}, q_{\mathrm{AFM}})$ reading emits exactly at the magnon frequency (0.86 THz measured), inherits the narrow magnon linewidth along ω_t, and softens toward the spin reorientation; the $2E_{12}$ harmonic reading sits rigidly at 1.00 THz, is broader, and grows one field power faster.

If the experimental buildup over tens of ps is real, it must be fed by the one reservoir that outlives every T_2: the CEF *populations*. A pulse-deposited excited population n_2 relaxes on T_1, and through the population channel of the same exchange family ($W_3\, S_\alpha S_\beta\, \lambda^3$, implemented in the solver but not part of the present minimal Hamiltonian) it shifts the effective Fe anisotropy as it relaxes—precisely the mechanism of thermally driven spin reorientation in $TmFeO_3$. This route makes a sharp experimental discriminator: a population-fed buildup is a *non-oscillatory* background (or a slow shift of the magnon frequency/equilibrium) whose growth time directly measures T_1, whereas growth of the 0.90 THz *oscillation amplitude* over tens of ps would require a persistent coherent source that the measured T_2's exclude—observing it would falsify the linewidth-constrained model class itself.

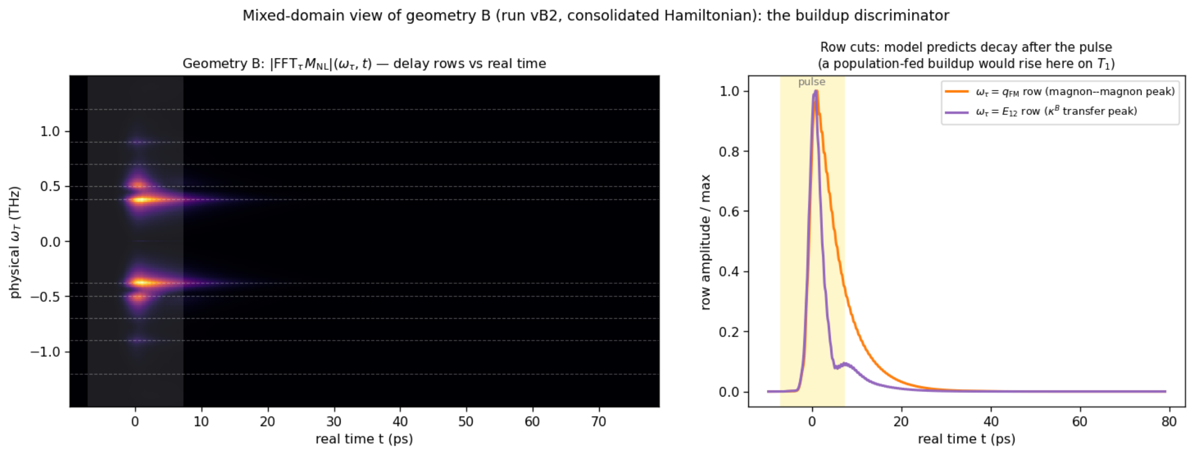

Figure S3: The buildup discriminator in the representation directly comparable to experiment: Fourier transform of $M_{\mathrm{NL}}(\tau, t)$ along the delay τ *only*, keeping the real detection time t (geometry B, run `vB2`). Left: $|\mathrm{FFT}_\tau|(\omega_\tau, t)$ map—each delay-frequency row evolves in real time. Right: cuts of the $\omega_\tau = q_{\mathrm{FM}}$ (magnon–magnon) and $\omega_\tau = E_{12}$ (κ^B-transfer) rows. The linewidth-constrained model predicts both rows peak with the pulse and decay within ∼20–30 ps; an experimental version of this plot in which the E_{12} row *rises* over tens of ps is the signature of the population-fed (T_1) mechanism.

Why the static vertex cannot replace the gated one. $W_1^{xz} S_x S_z \lambda^1$ is trilinear; around the Γ_2 order it decomposes into a static CEF shift, two energy-conserving *bilinear* hops ($F_x\, \delta S_z\, \lambda^1$ and $G_z\, \delta S_x\, \lambda^1$: pure hybridization), and a genuine two-fluctuation converter $\delta S_x\, \delta S_z\, \lambda^1$. The converter piece is real and retained—it drives the post-pulse buildup through the nearly closed triad $q_{\mathrm{FM}} + E_{12} = 0.88 \approx q_{\mathrm{AFM}}$ (0.02 THz detuning)—but it is fluctuation-fed (its energy leg is a probe-launched magnon, one factor of drive×susceptibility weaker and spectrally narrow) and detuning-limited (secular accumulation over tens of ps), so it produces an *envelope*, not an impulsive phase-stamped kick. At $W_1^{xz} = 0.01$ the isolated run has sizable absolute amplitude at $(+0.5, 0.90)$ (0.72 of map max) but it is the *tail* of the magnon–magnon peak—no genuine local maximum on the E_{12} row.

A careful strength sweep in the consolidated configuration ($W_1^{xz} \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.09\}$, $\kappa^B = 0$; runs `vBwxz*`) shows that raising the static vertex *never* creates a second resolved peak. Instead the two delay rows coalesce—the leading peak's delay phase migrates continuously, $0.38 \to 0.40 \to 0.48 \to 0.47$ THz across the sweep, the resolved $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})/(E_{12}, q_{\mathrm{AFM}})$ doublet structure never appears at any strength, and the hybridization-shifted E_{12} branch leaks into the M1 channel as emission strays at $\omega_t = 0.44$–0.45 THz (growing to 0.15–0.16 by $W_1^{xz} = 0.09$). This is the structural signature of a static mixer: it *moves* peaks (level repulsion of the hybridized normal modes—also shifting the mode frequencies off their 1D-measured values), whereas the experiment shows two resolved peaks at the *unshifted* delay energies 0.38 and 0.50 THz. A gated converter *adds* a peak while leaving the normal modes untouched; with $\kappa^B_{1y} = 0.0035$ the census gives $(+0.38, 0.90) = 1.00$ and a separate $(+0.53, 0.89) = 0.46$, both on the measured lines. The necessity of the field-gated vertex therefore does not rest on any operating-point fine-tuning: static mixing and gated conversion are qualitatively different operations on the 2D map. In geometry B the Tm CEF channels $\lambda^{4\text{–}7}$ are *exactly* zero (machine precision): the $E \parallel a, H \parallel c$ drive closes on the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra, an SU(2) closure that protects geometry B from CEF contamination.

Finite temperature (regenerated final-scenario runs `vB2`): $(q_{\rm FM}, q_{\rm AFM})$ is a pure two-magnon process and temperature-flat; $(E_{12}, q_{\rm AFM}) \propto n_1 - n_2$ falls 0.46 (0 K) → 0.45 (5) → 0.40 (8) → 0.36 (10 K). Both peaks remain resolved through ~ 10 K, defining the working window 5–8 K.

S8 The same-polarization channel: pathways, exact single-ion verification, and the falsification ladder

S8.1 Configuration

The final same-polarization configuration (run `it14hr`; parameter files in `tmfeo3_2dcs_final/params_samepol_fina` is the unchanged v4 Hamiltonian with: measured pulse; drive vector $(0, 3.0, 0, 0, \mathbf{0}, 0, 0.9128, 0)$ on $(\lambda^1, \ldots, \lambda^8)$—i.e. the $1 \to 3$ leg removed ($\mu^z_{13} = 0$ in drive); per-leg amplitude twice the weak-probe baseline; dampings of Table S2 plus population relaxation $\gamma(\lambda^{3,8}) = 0.08$; Boltzmann ensemble; detection composite $0.05\,\lambda^4 + 4.4\,\lambda^6$ ($\mu^z_{13}/\mu^z_{23} \approx 1\%$ in emission), magnon M1 excluded per Table S2.

S8.2 Pathway assignments

With $\mu^z_{13} = 0$ the Liouville pathways of the five listed peaks are:

peak (ω_T, ω_t)	model (10 K)	side	pathway
$(+0.49, 0.72)$	1.00	NR	$\rho_{12} \xrightarrow{\text{probe } 1\leftrightarrow 2} n_2(\tau\text{-phase}) \xrightarrow{\text{probe } 2\leftrightarrow 3} \rho_{23}$
(0.5, 0)	sharpest $S_{\rm DC}$ line	—	same storage, zero-frequency output leg
$(+0.49, 1.20)$	0.78	NR	$\rho_{12} \xrightarrow{\text{probe } 2\leftrightarrow 3} \rho_{13}$, emission throttled by μ^z_{13}
$(+0.91, 1.18)$	$0.20 \to 0.51$ ($10 \to 20$ K)	NR	$q_{\rm AFM}$ delay, Fe–Tm handoff, thermally assisted
(0.9, 0.38)	absent	—	carrier exists (b-axis M1 $q_{\rm FM}$ ridge); no $q_{\rm AFM} \to q_{\rm FM}$ vertex
$(-0.47, 0.70)$	0.55	R	rephasing remnant (exact single-ion floor; prediction)
$(+0.99, 0.71)$	0.49	NR	inter-site $2E_{12}$ two-quantum line (prediction)

S8.3 Exact single-ion verification

A standalone integrator of the 8-dimensional $\vec{\lambda}$ equation (same CEF energies, dampings, measured pulse, same census pipeline; ~ 13 s per 2D map versus ~ 10 min for the lattice) reproduces the lattice 0.5-family census exactly, proving the family is single-ion physics and enabling exhaustive scans. Two facts emerge: (i) the rephasing/non-rephasing ratio of the (E_{12}, E_{23}) transfer peak has a floor of 0.47–0.58 under this pulse across all drive strengths, dipole-phase choices, and dampings—the rephasing remnant is an exact prediction of any three-level ion driven by this waveform; (ii) the $\omega_\tau = 2E_{12}$ two-quantum line is absent in the exact single-ion dynamics and is therefore an inter-site (mean-field) product.

S8.4 Falsification ladder

The dark-dipole scenario was reached by exhausting the alternatives; each row was run in the full lattice model with the blind census:

hypothesis	run	outcome
static mixing field $h_{\lambda^6} = 0.3$	it8	over-strong: transition lines shift off calibration
perturbative $h_{\lambda^6} = 0.06$	it9	pumps $(q_{\mathrm{AFM}}, E_{13})$ to 0.9 (wrong direction); side unchanged
CEF detuning 0.02 THz (ladder break)	it10	interloper survives; larger detuning barred by the observed degenerate lines
dipole triple-product sign flip	it11	census unchanged
dark μ_{13} at low drive	it12	population route floods the $(0, E_{23})$ band; needs the $2\times$ drive + T_1 combin
dark μ_{13}, $2\times$ drive, T_1	**it14**	**adopted**: both transfer peaks NR, correct hierarchy
negative μ_{13} leg, lower drive	it15	interloper returns ($\lvert\mu_{13}\rvert$ acts sign-independently); magnon-delay family floo

The $(0.9, 0.38)$ peak remains the one open item: the q_{FM} emission carrier exists as a uniformly τ-modulated b-axis M1 ridge in the Fe S_y channel (Γ_2 puts $\delta F_{y,z}$ at q_{FM}), but the model lacks a $q_{\mathrm{AFM}} \to q_{\mathrm{FM}}$ transfer vertex of sufficient strength; one probed configuration (it15) produced a genuine $(+0.89, 0.33)$ maximum at 0.21 relative, evidence the peak is reachable in an extended Hamiltonian class.

S9 The consolidated configuration and its verified censuses

Every spectrum in the paper is produced by *one* Hamiltonian and *one* drive model; only the field geometry changes between channels. Couplings (code units): Fe exchange/anisotropy/DM of the calibrated orthoferrite model ($D_1 = 0.049$, $D_2 = 0.014$); Tm CEF $e_1 = 2.0678$, $e_2 = 4.9628$ meV; Fe–Tm $W_1^{yy} = 0.01$, $W_1^{xz} = 0.01$, $\kappa_{1y}^{B} = 0.0035$; drive legs $(d_z, \mu_{13}^z, \mu_{23}^z) \propto (3.0,\ 0,\ 0.9128)$ for $E \parallel c$ (the dark-μ_{13} rule) and the auto-projected Zeeman drive for the magnetic field; dampings and emission weights from Table S2. The final regeneration (runs `vA`, `vB2`, `it14hr`; three Boltzmann seeds each) gives the blind censuses, normalized per channel [amp, (ω_T, ω_t), $+ =$ non-rephasing]:

channel	leading genuine peaks (T=0)	reading
A cross (λ^2)	1.00 $(+0.91, 0.50)$; 0.46 $(-0.90, 0.50)$; 0.24 $(+0.90, 0.89)$	W_1^{yy} peak; R partner; magnon shoulder
B cross (S_x)	1.00 $(+0.38, 0.90)$; 0.46 $(+0.53, 0.89)$; 0.17 $(-0.42, 0.90)$	magnon–magnon; κ^B transfer; R partner
A same (composite)	1.00 $(+0.49, 0.71)$[1]; 0.71 $(+0.49, 1.20)$; 0.49 $(-0.47, 0.71)$	population-route transfer; μ_{13}-throttled; R

The complete censuses at all temperatures, together with the normalized spectra themselves, are archived as `paper/data/census_final_scenario.json` and `paper/data/*.npz`.

S10 Data and code availability

The paper folder is self-contained: `paper/params/` holds the exact `.param` files of every final-scenario run (`vA`, `vB2`, `it14hr`; three thermal seeds each), `paper/data/` the distilled spectra and censuses, `paper/scripts/` the census pipeline and the standalone qutrit integrator. The long-form derivation note (`tmfeo3_foundation.tex`: transported-gauge orbit formulas, vertex diagrams, complete

[1] Cross peaks normalized to the leading cross peak; the $\omega_T \approx 0$ pump–probe bands are excluded per the census convention of Sec. S4.

written-out Hamiltonian, mode inventory) is archived alongside (`tfo_project/`); every spectrum in the paper can be regenerated from the committed `.param` files with the public `ClassicalSpin_Cpp` solver.